Functions (Maps) Lecture N°4

1 Lic Data Science

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Functions or maps
Definitions-The graphic
Operations on maps
Direct image-The preimage
Injection, surjection and bijection
One-to-one functions and onto functions
Composition and inverse maps

Summary

Functions or maps

- Definitions-The graphic
- Operations on maps
- Direct image-The preimage

Injection, surjection and bijection

- One-to-one functions and onto functions

Composition and inverse maps

Definition

Let E and F be two sets. A map f from E into F is defined as a triplet $(E; F; G_f)$, where G_f is a subset of $E \times F$ satisfying the following property:

$$\forall x \in E; \exists! y \in F : (x; y) \in G_f.$$

- The set E is called **the domain** (or source) of f.
- The set F is called **the codomain** (or target) of f.
- The element y is called **the image** of x under f and is expressed as $y = f(x)$.
- The element x is called **the preimage** or antecedent of y under f.
- We can represent the map f using the notation:

$$\begin{array}{rccc} f: & E & \to & F \\ & x & \to & y = f(x) \end{array}$$

- This notation indicates that f maps elements from the set E to elements in the set F, and for each $x \in E$, the image $f(x)$ is uniquely determined in F.
- $f(x)$ is the function term, x is called the argument of f.
- Other names: function, transformation
- We denote by E^F the set of all maps from E into F.

Example

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- For any set X the following map

$$\begin{array}{rccc} id_X : & X & \to & X \\ & x & \mapsto & x \end{array}$$

is called identity map of X.

Example

- Let $A \subset X$. The characteristic (or indicator) map of A,

$$\chi_A : X \to 0, 1,$$

is defined by

$$\chi(x) = \begin{cases} 1 & if \quad x \in A \\ 0 & if \quad x \notin A. \end{cases}$$

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- The square function $f : \mathbb{N} \to \mathbb{N}$ is defined by $f(n) = n^2$, which we also write as $f : n \mapsto n^2$.

- Let S and T be two sets. For a fixed element a in T: The map

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is called **constant**. This map sends every element of S to the same element a in T.

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- Let X and A be two sets such that $A \subseteq X$. The map

$$\begin{array}{rccc} i_A: & A & \to & X \\ & x & \mapsto & x \end{array}$$

is called **the canonical injection**. This map embeds A into X by mapping each element of A to itself in X.

- Let E and F be two sets. The map

$$\begin{array}{rccc} \rho_S : & S \times T & \to & S \\ & (x, y) & \mapsto & x \end{array}$$

is called **the projection map** over S. In the same way the map

$$\begin{array}{rccc} \rho_T : & S \times T & \to & T \\ & (x, y) & \mapsto & y \end{array}$$

is called **the projection map over** T. These maps extract the first and second components, respectively, from the pairs in $S \times T$.

The graphic

Definition

The graphic of a function f is the set of all points in the plane of the form $(x, f(x))$. We could also define the graph of f to be the graph of the equation $y = f(x)$. So, the graph of a function if a special case of the graph of an equation.

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$$G(f) = \{(x, f(x)) \in E \times F \,|\, x \in E\} \subset E \times F$$

The graphic

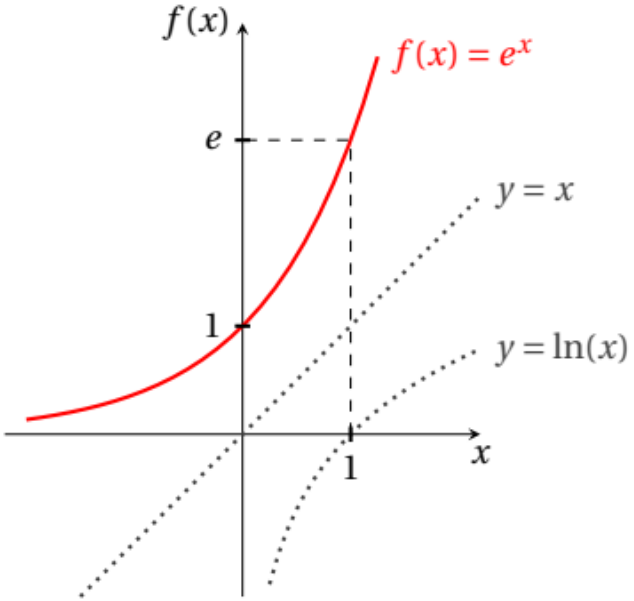
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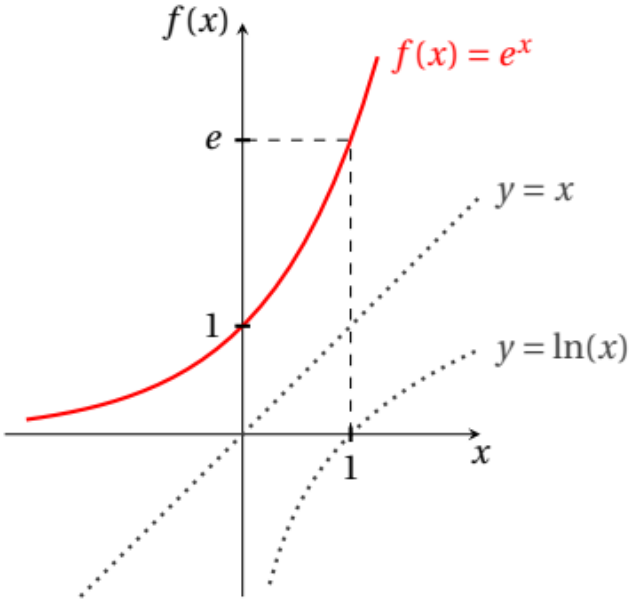
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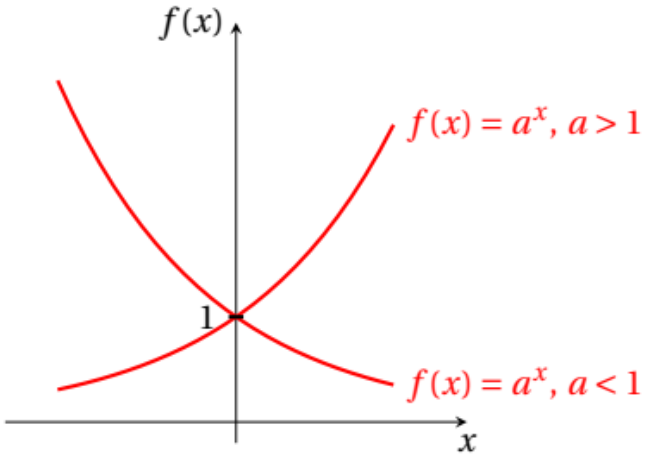
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Example

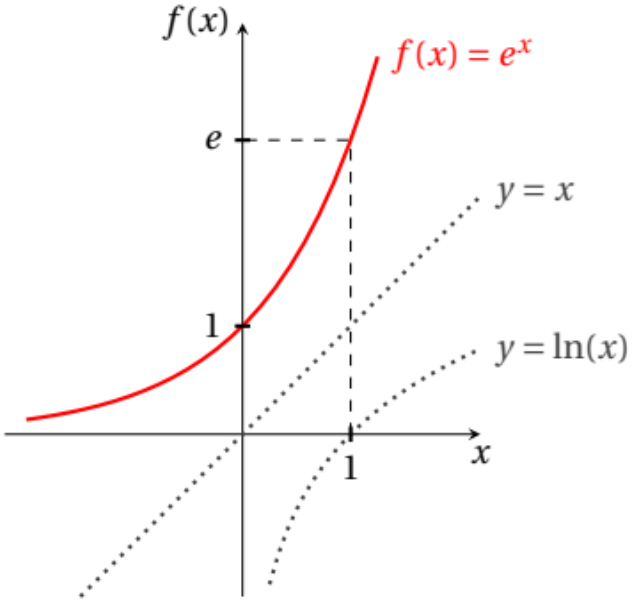


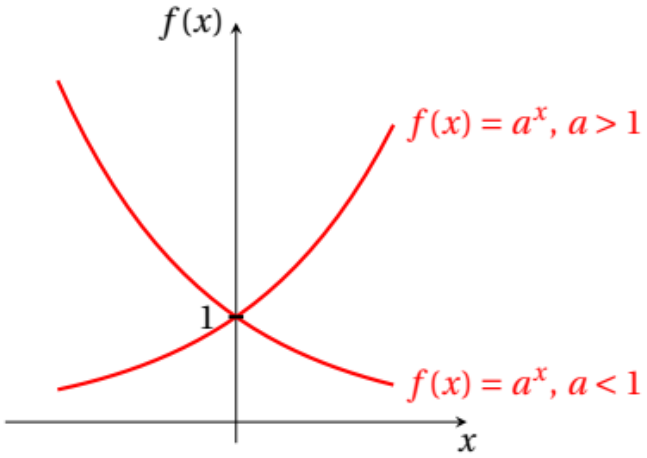
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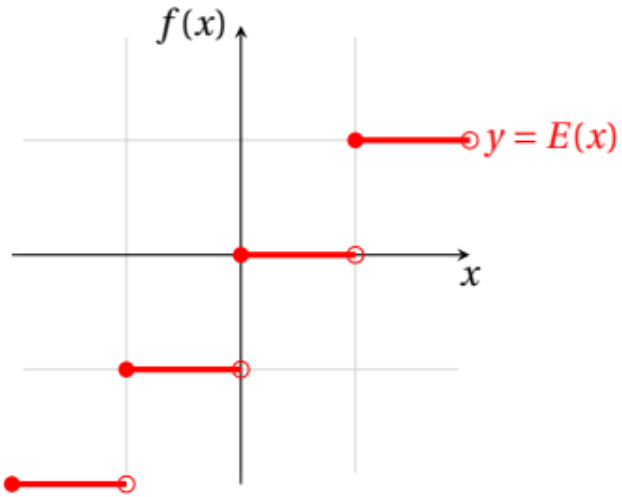




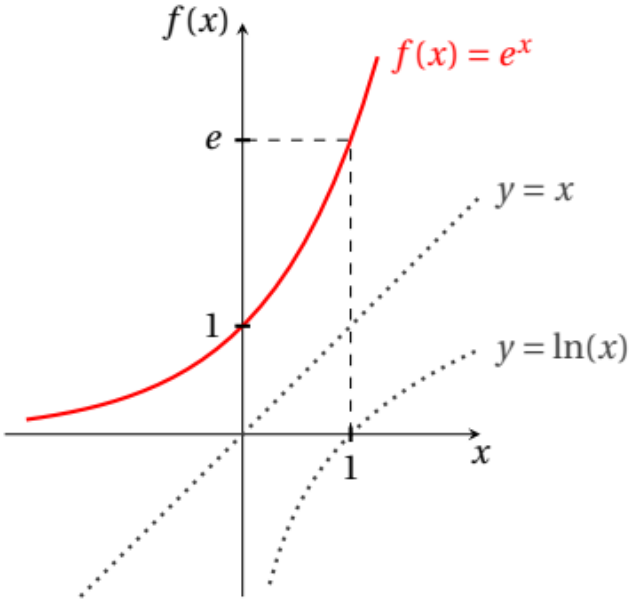
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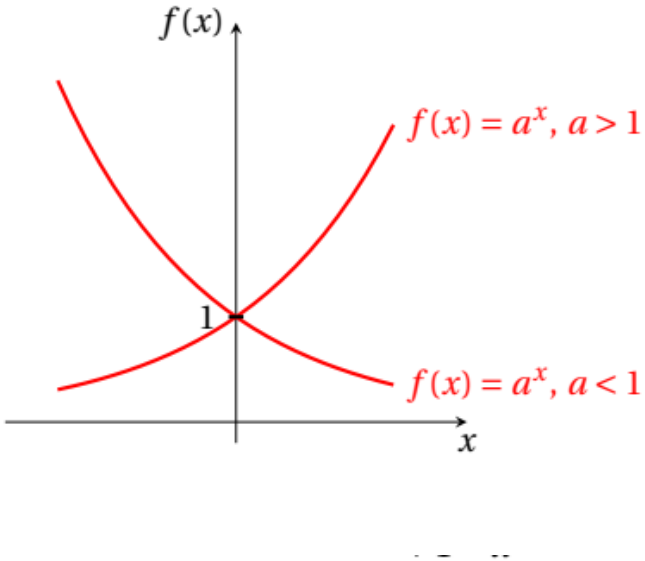


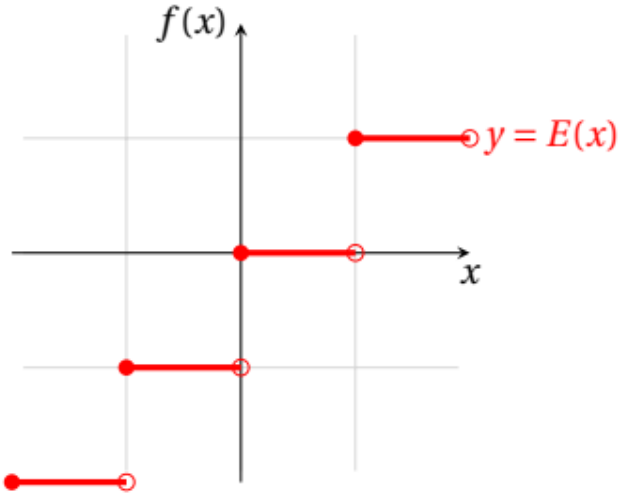


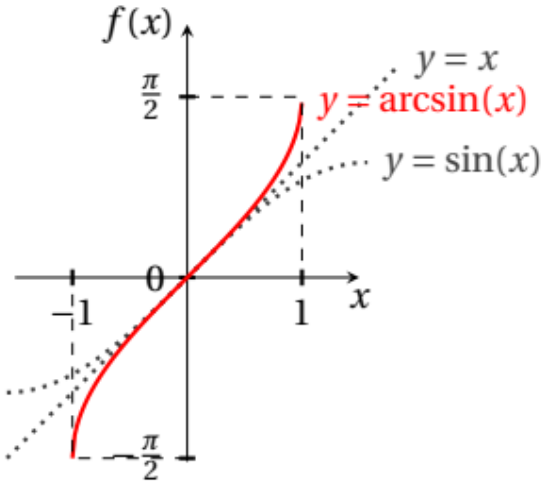


Example


f(x)
f(x) = e^x
e
y = x
1
y = ln(x)
1
x


f(x)
f(x) = a^x, a > 1
1
f(x) = a^x, a < 1
x


f(x)
y = E(x)
x


f(x)
y = x
π/2
y = arcsin(x)
y = sin(x)
0
−1
1
x
−π/2

The equality of applications

Two maps

$$\begin{array}{rccc} f: & X & \to & Y \\ & x & \mapsto & f(x) \end{array}$$

and

$$\begin{array}{rccc} g: & X' & \to & Y' \\ & x & \mapsto & g(x) \end{array}$$

are called **equal** if $X = X'$, $Y = Y'$ and

$$\forall x \in X; f(x) = g(x).$$

We then note $f = g$.

Example

The following three maps are **not equal**

$$\begin{array}{rccc} f: & \mathbb{R}_+ & \to & \mathbb{R}_+ \\ & x & \mapsto & x^2, \end{array}$$

$$\begin{array}{rccc} g: & \mathbb{R}_- & \to & \mathbb{R} \\ & x & \mapsto & x^2, \end{array}$$

$$\begin{array}{rccc} h: & \mathbb{R} & \to & \mathbb{R} \\ & x & \mapsto & x^2, \end{array}$$

because, even the three formulas are equal, **the sources** and **the targets** are **not equal**. But there are relations between them, the following notions clarify this

Operations on maps

Definition

Let $f : E \to F$ be a function, and let A be a subset of E. Then the function,

$$g : A \to F,$$

which is defined by $g(x) = f(x), \forall x \in A$, is called **the *restriction*** of f to A. We denote this restriction g by

$$f_{|A}.$$

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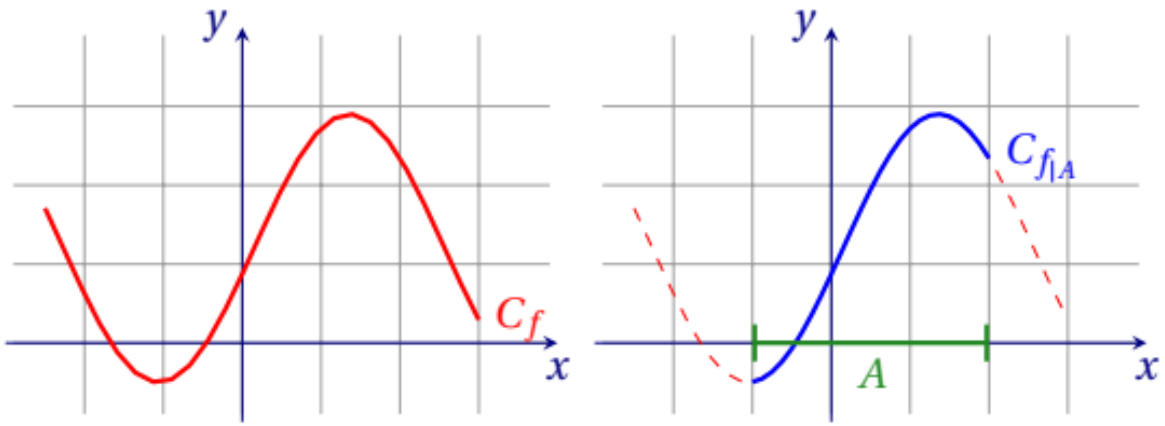
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Example

Let f and h be the following maps:

$$\begin{array}{rccc} f: & \mathbb{R} & \to & \mathbb{R} \\ & x & \mapsto & |x+1|, \end{array}$$

and

$$\begin{array}{rccc} h: & \mathbb{R} & \to & \mathbb{R} \\ & x & \mapsto & x+1, \end{array}$$

h is **the restriction** of f to $]-1,+\infty[$

Definition

Let $f : E \to F$ be a function, and let $E \subseteq E'$. Then the function,

$$g : E' \to F,$$

is called ***an extension*** of f from E to E', if

$$g(x) = f(x), \forall x \in A,$$

that is $g_{|A} = f$. i.e: $\forall x \in E; f(x) = g(x)$.

Example

- Let f and g be the following maps:

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$$\begin{array}{rccc} g: & \mathbb{R} & \longrightarrow & \mathbb{R} \\ & x & \mapsto & \begin{cases} x & \text{if} \quad x \geq 0 \\ 1 & \text{if} \quad x < 0 \end{cases} \end{array}$$

  The map g is an extension of f to $\mathbb{R}$.

- Let $f : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ the function defined by $f(x) = \frac{\sin x}{x}$.

Example

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- Let $f : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ the function defined by $f(x) = \frac{\sin x}{x}$.Its representative curve strongly suggests extending it to integer $\mathbb{R}$ by setting $g(0) = 1$.

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- Let $f : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ the function defined by $f(x) = \frac{\sin x}{x}$.Its representative curve strongly suggests extending it to integer $\mathbb{R}$ by setting $g(0) = 1$. Why ?

Definition

A function $f : A \to B$ is called **a numerical function**, if B is a set of numbers.

Majority of the exercises and examples above are numerical functions

Direct image

Definition

The direct image of A by f; denoted $f(A)$, is the subset of F; defined by:

$$f(A) = \{f(x) \,|\, x \in A\}$$

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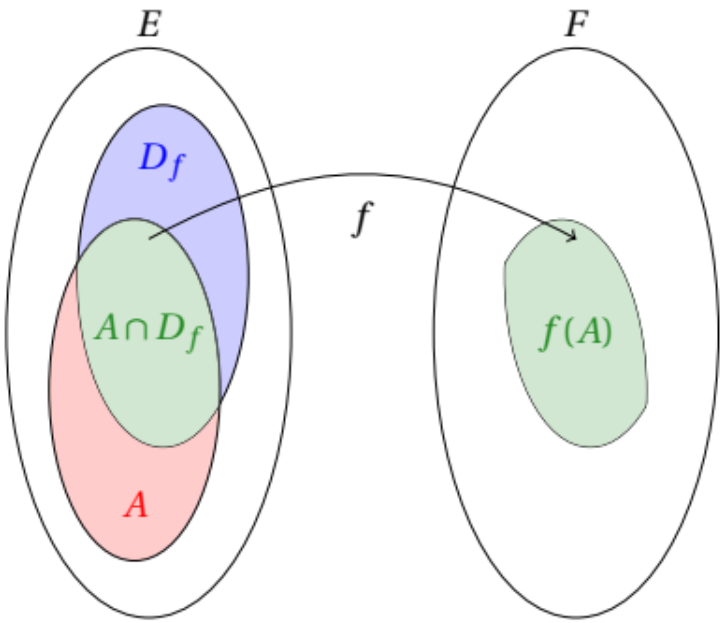
$$f(A) = \{f(x) \,|\, x \in A\} = \{y \in F \,|\, (\exists x \in A) : (y = f(x))\}.$$

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Example

Let f de $\mathbb{R}$ dans $\mathbb{R}$ defined by $f(x) = |x|$. Then

- $f(\{-1, 2\}) = \{f(-1), f(2)\} = \{1, 2\}$
- $f([-3, -1]) = \{f(x) / \quad x \in [-3, -1]\}$,we have $-3 \leq x \leq -1 \Rightarrow 1 \leq |x| \leq 3$ donc $1 \leq f(x) \leq 3$ Thus $f([-3, -1]) = [1, 3]$.

The preimage

Definition

Inverse image;The preimage Let $f : E \to F$ be a function, and let $B \subset F$. **The preimage** of B by f ,denoted $f^{-1}(B)$; is the subset of S defined by:

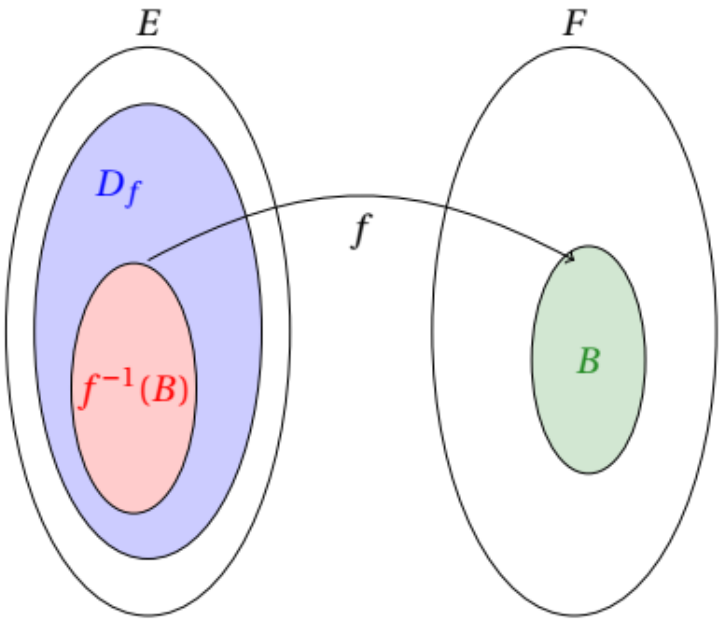
$$f^{-1}(B) = \{x \in E \mid f(x) \in B\}.$$

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Remark

Note that, even though the inverse images are expressed under the symbol f^{-1}, this symbol does not mean the inverse function of f . It is simply used pure symbolically. Even if f is not invertible, one can define and write the corresponding inverse images as above.

Example

Let $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^2$. Find

$$f^{-1}(\{4\}), f^{-1}(\{-1\}), f^{-1}([-1, 4]).$$

1. $f^{-1}(\{4\}) = \{x \in \mathbb{R}/f(x) = 4\} = \{x \in \mathbb{R}/x^2 = 4\} = \{-2, 2\}$.
2. $f^{-1}(\{-1\}) = \{x \in \mathbb{R}/f(x) = -1\} = \{x \in \mathbb{R}/x^2 = -1\} = \emptyset$.
3. $f^{-1}([-1, 4]) = \{x \in \mathbb{R}/ -1 \leq f(x) \leq 4\} = \{x \in \mathbb{R}/ -1 \leq x^2 \leq 4\} = [-2, 2]$.

Proposition

Let $f : E \to F$ be a function and A, B be subsets of E. Then

1. $A \subseteq B \Rightarrow f(A) \subseteq f(B)$,
2. $f(A \cup B) = f(A) \cup f(B)$,
3. $f(A \cap B) \subseteq f(A) \cap f(B)$,
4. $f(A \setminus B) \supseteq f(A) \setminus f(B)$.

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Proof.

Let $A \subseteq B$. Since $y \in f(A)$ is equivalent to $y = f(x)$ for some x in A, and $A \subseteq B$ we have $y = f(x)$ and $x \in B$, then $y \in f(B)$. □

Proof [2: Worksheet N°:12]
2) Let $y \in f(A \cap B)$ implies $y = f(x)$ with $x \in A \cap B$. This leads to $y = f(x)$ with $x \in A$ and $x \in B$. Then $y \in f(A)$ and $y \in f(B)$. Therefore

$$f(A \cap B) \subset f(A) \cap f(B).$$

The statement $f(A) \cap f(B) \subseteq f(A \cap B)$ is false and can be disproven by a simple counterexample involving a non-injective function. For instance, let the function be $f(x) = 0$ for all x, with the domain $X = \{0, 1\}$ and subsets $A = \{0\}$ and $B = \{1\}$. Here, $f(A) = \{0\}$, $f(B) = \{0\}$, and $f(A) \cap f(B) = \{0\}$. However, $A \cap B = \emptyset$, so $f(A \cap B) = f(\emptyset) = \emptyset$. Since $\{0\} \not\subseteq \emptyset$, the statement is false.

3) We have $y \in f(A \cup B)$ is equivalent to $\exists x \in A \cup B : y = f(x); x \in A \cup B$, which leads to $x \in A$ or $x \in B$ and therefore

$$f(x) \in f(A) or f(x) \in f(B),$$

which means that $y \in f(A) \cup f(B)$. Then

$$f(A \cup B) \subset f(A) \cup f(B). \tag{1}$$

Conversely, $A \subseteq A \cup B$ gives $f(A) \subseteq f(A \cup B)$ and $B \subseteq A \cup B$ gives $f(B) \subseteq f(A \cup B)$. Then

$$f(A) \cup f(B) \subseteq f(A \cup B) \tag{2}$$

The inclusions (1) and (2) give the desired equality.

Proposition

Let $f : E \to F$ be a function and A, B be subsets of F. Then

1. $A \subseteq B \Rightarrow f^{-1}(A) \subseteq f^{-1}(B)$,
2. $f^{-1}(A \cup B) = f^{-1}(A) \cup f^{-1}(B)$,
3. $f^{-1}(A \cap B) = f^{-1}(A) \cap f^{-1}(B)$,
4. $f^{-1}(A - B) = f^{-1}(A) - f^{-1}(B)$.

Proof.

1. Assume $A \subseteq B$, For any $x \in E$; $x \in f^{-1}(A)$ is equivalent to $f(x) \in A$. Since $A \subseteq B$, we have $f(x) \in B$, equivalently $x \in f^{-1}(B)$. Thus,

$$x \in f^{-1}(A) \subseteq x \in f^{-1}(B).$$

2. For any $x \in F, x \in f^{-1}(A \cup B)$ is equivalent to $f(x) \in A \cup B$, that is $f(x) \in A$ or $f(x) \in B$ so $x \in f^{-1}(A)$ or $x \in f^{-1}(B)$, which is equivalent to

$$x \in (f^{-1}(A) \cup f^{-1}(B)).$$

   This prroves that the equality in 2) holds.

3. It can be proven in the same way as 2)

□

4. If $x \in f^{-1}(A-B)$, then $f(x) \in (A-B)$, that is, $f(x) \in A$ and $f(x) \notin B$, which is equivalent to $x \in f^{-1}(A)$ and $x \notin f^{-1}(B)$ i.e., $x \in f^{-1}(A) - f^{-1}(B)$. Then the equality

$$f^{-1}(A-B) = f^{-1}(A) - f^{-1}(B).$$

Injections

Definition

Let $f : E \to F$ be a function. Then f is called a **one-to-one function**, if

$$\forall x, y \in E, x \neq y \Rightarrow f(x) \neq f(y),$$

Or, $f(x) = f(y) \Rightarrow x = y$.

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$$\forall x, y \in E, x \neq y \Rightarrow f(x) \neq f(y),$$

Or, $f(x) = f(y) \Rightarrow x = y$.A one-to-one function is also called an injection.

Example

1. $f : \mathbb{N} \to \mathbb{N}, n \mapsto n+1$ is one-to-one function
   if $n = n' \Rightarrow n+1 = n'+1$ then $f(n) = f(n')$
2. $f : \mathbb{R} \to \mathbb{R}, x \mapsto x^2$ is not a one-to-one function
   $-2 \neq 2$ but $f(-2) = f(2) = 4$.
3. $f : \mathbb{R} \to \mathbb{R}_+, x \mapsto x^2$ is one-to-one function.

Surjective

Definition

Let $f : E \to F$ be a function. We say that
" f is a function from E onto F " or " f maps E onto F",
or " f is an onto function,"
if

$$Rng(f) = F.$$

In other words, " f is onto function" if and only if

$$\forall y \in F,$$

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In other words, " f is onto function" if and only if

$$\forall y \in F, \exists x \in E / f(x) = y.$$

An onto function is also called a surjection.

Example

The function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = \frac{2x}{1+x^2}$ is not an onto function
since if $y = 2$ the equation $f(x) = 2 \Leftrightarrow x = 2(1 + x^2)$then $x^2 - x + 1 = 0$ which has no real solutions

Bijection

Definition

A function $f : E \to F$, which is both one-to-one and onto is called a bijection, or an one–one correspondence.

$$\forall y \in F,$$

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Example

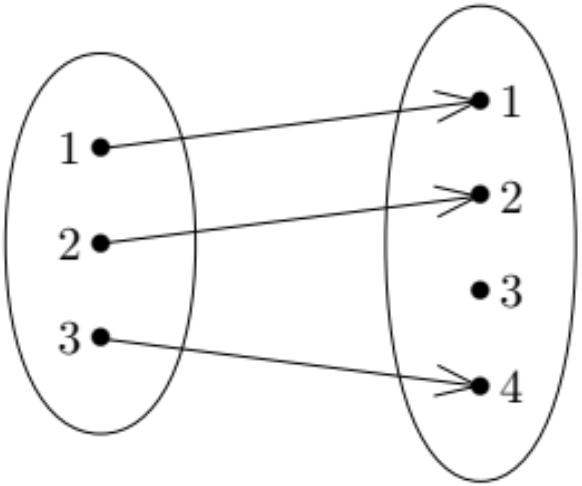
Let

$$A = \{2n + 1 | n \ \ is \ \ an \ \ integer\},$$

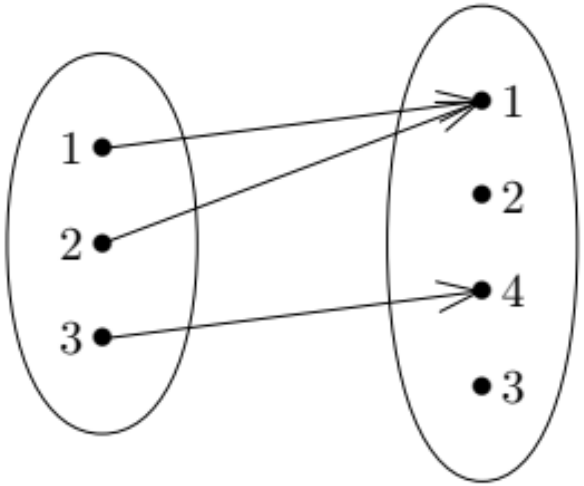
and

$$B = \{2n | n \ \ is \ \ an \ \ integer\},$$

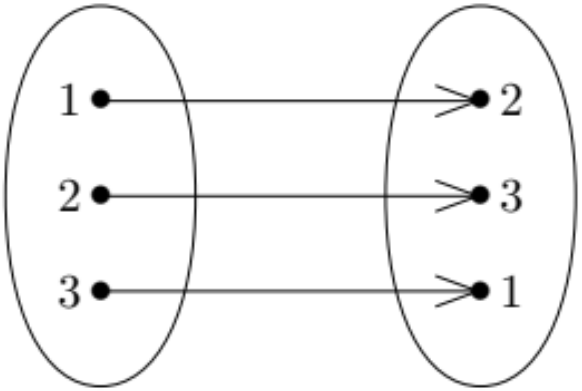
and let $f(x) = 2x$, and $g(x) = x + 1$. The function $f : A \to B$ is not a bijection, because it is not onto, meanwhile, the function $g : A \to B$is a bijection, since it is both one-to-one and onto.



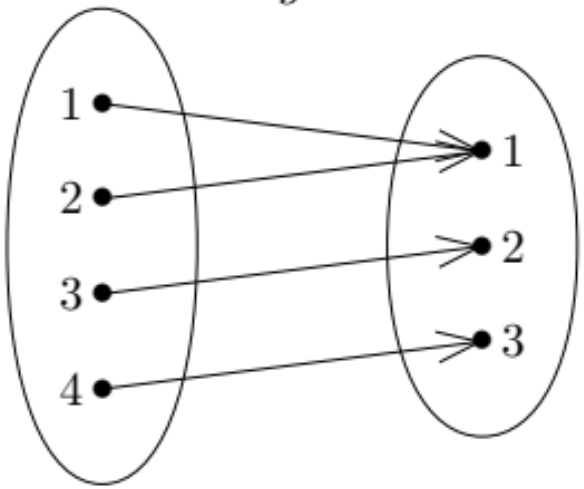
f



g



h



k

1. The mapping f is injective but not surjective,
2. g is neither injective nor surjective,
3. h is bijective, and
4. k is surjective but not injective.

The Composition

Definition

Let $f : E \to F$ and $g : F \to G$ be functions. The composition of f and g taken in this order, and denoted $g \circ f$, is the map defined by

$$\begin{array}{rccc} g \circ f & E & \to & G \\ & x & \mapsto & g\left(f(x)\right) \end{array}.$$

And the construction of composite functions, or the operation $(\circ)$ is called the composition.i.e,

$$(g \circ f)(x)$$

Functions (Maps) Lecture N°4

A. DJOUMAKH

Functions or maps

Definitions-The graphic

Operations on maps

Direct image-The preimage

Injection, surjection and bijection

One-to-one functions and onto functions

Composition and inverse maps

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$$(g \circ f)(x) = g(f(x)), \forall x \in E.$$

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Example

Let $f : \mathbb{R}_+ \to \mathbb{R}$ be the function defined by $f(x) = \sqrt{x}$ and $g : \mathbb{R} \to \mathbb{R}$ defined by $g(y) = \sin y$.

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The above example demonstrates that the composite functions $f \circ g$ and $g \circ f$ are not identical, in general, and hence, the composition $(\circ)$ is not commutative (as an operation on functions).

The inverse function

Definition

Let $f : E \to F$ be a bijective function. The map denoted by f^{-1}, and defined from F into E which associates to any $y \in F$ the unique element $x \in E$ such that $y = f(x)$, is called **the inverse** map of f.

$$\begin{array}{cccc} f^{-1} & F & \to & E \\ & x & \mapsto & f^{-1}(x) \end{array}$$

Thus, we have:

$$(f(x) = y) \Leftrightarrow (x = f^{-1}(y)).$$

where $x \in E$ and $y \in F$.

Remark

- Let f be a bijective map from E into F. Note that we have

$$E \xrightarrow{f} F \xrightarrow{f^{-1}} E$$

so

$$x \xrightarrow{f} y = f(x) \xrightarrow{f^{-1}} f^{-1}(y) = x,$$

therefore

$$(\forall x \in E) \left(f^{-1} \circ f\right)(x) = x$$

that is, $f^{-1} \circ f$ is the identity map on E i.e., $f^{-1} \circ f = id_E$. In a similar way, we have $f \circ f^{-1} = id_F$.

- Let f be a map from E into F. For any $y \in F$ we have

$$f^{-1}(\{y\}) = \{x \in E : f(x) = y\}.$$

Then $f^{-1}(\{y\})$ is a subset of E and is defined even f is not bijective. But $f^{-1}(y)$ is an element of E and is defined only when f is bijective and in this case we have

$$f^{-1}(\{y\}) = \{f^{-1}(y)\}.$$

Example

Let the function $f : [1, +\infty[\to [0, +\infty[$ telle que $f(x) = x^2 - 1$.
we have

i. f is one-to-one:
$f(x) = f(y) \Rightarrow x^2 - 1 = y^2 - 1 \Rightarrow x = \pm y$ where $x, y \in [1, +\infty[$ then x, y have the same sign $\Rightarrow x = y$.

ii. f is onto : let $y \in [0, \infty[$. We are looking for an element $x \in [1, +\infty[$ such that $y = f(x) = x^2 - 1$.Then $x = \sqrt{y+1}$

ii. f is therefore bijective and its inverse function is $f^{-1} : [0, +\infty[\to [1, +\infty[$ defined by $f^{-1}(y) = \sqrt{y+1}$.

Proposition

Let $f : E \to F$ and $g : F \to G$ be two maps.

1. *If f and g are injective, then $g \circ f$ is injective.*
2. *If f and g are surjective, then $g \circ f$ is surjective:*
3. *If f and g are bijective then, $g \circ f$ is bijective and $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$*

Proof:

1. For any $x, x' \in E$, we have $(g \circ f)(x) = (g \circ f)(x')$ is equivalent to $g(f(x)) = g(f(x'))$. This gives $f(x) = f(x')$ because g is injective. Then $x = x'$ because f is injective. Therefore $g \circ f$ is injective.

2. Let $z \in G/$ Since g is surjective, there exists $y \in F$ such that $z = g(y)$. On the other hand, f is surjective, so there exists $x \in E$ such that $y = f(x)$, which gives $z = g(y) = g(f(x)) = (g \circ f)(x)$. Therefore, $g \circ f$ is surjective.

The bijectivity of gf follows from (1) and (2). On the other hand, for all $z \in G$. We have

$$(g \circ f)^{-1}(z) = x \Leftrightarrow z = (g \circ f)(x),$$

but we have

$$(f^{-1} \circ g^{-1})(z) = f^{-1}\left(g^{-1}(z)\right) = f^{-1}(y)$$

(with $z = g(y) = x$ and $y = f(x)$), where

$$\forall z \in G : (f^{-1} \circ g^{-1})(z) = (g \circ f)^{-1}(z),$$

which means that

$$(g \circ f)^{-1} = f^{-1} \circ g^{-1}.$$

Proposition

Let E and F be two sets, and let $E \xrightarrow{f} F$ and $F \xrightarrow{g} E$ be two maps such that

$$g \circ f = id_E \tag{3}$$

and

$$f \circ g = id_F \tag{4}$$

Then

$$f, g \text{ are bijective, and } g = f^{-1}.$$

Proof.

By (3), f is injective, g is surjective, and by (4), g is injective, and f is surjective. Therefore, f and g are bijective. Moreover, if y is an element of F, the equality $f(g(y)) = y$ is equivalent to $g(y) = f^{-1}(y)$, thus $g = f^{-1}$. Similarly, we see that $f = g^{-1}$. □